

63rd ASH Annual Meeting Abstracts

ORAL ABSTRACTS

704.CELLULAR IMMUNOTHERAPIES: CLINICAL

The Intestinal Microbiota Correlates with Response and Toxicity after CAR T Cell Therapy in Patients with B-Cell Malignancies

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Abstract Introduction: Cellular immunotherapy with CD19-targeted chimeric antigen receptor (CAR) T cells has provided new therapeutic options for patients with high-risk hematologic malignancies. Following this therapy, patients may experience disease relapse or CAR-mediated toxicity due to cytokine release syndrome (CRS) or immune effector cell-associated neurotoxicity syndrome (ICANS). Recent studies have confirmed that the intestinal microbiome can modulate the anti-tumor immune response to chemotherapy, immune checkpoint blockade, graft-versus-host disease after allogeneic hematopoietic cell transplantation, and adoptive cellular therapy. The contribution of the intestinal microbiome on the function of CAR T cells *in vivo* both with respect to their anti-tumor function and their propensity to induce toxicities is not known. Hence, in a multi-center study we analyzed the association between clinical outcomes and (1) antibiotic exposure prior to CAR T cell infusion and (2) the composition and diversity of the fecal microbiome.

Methods and Results: We retrospectively collected clinical data and antibiotic exposures from patients with acute lymphoblastic leukemia (ALL, n=91) and non-Hodgkin lymphoma (NHL, n=137) treated with investigational or commercial CD19 CAR T cells at Memorial Sloan Kettering Cancer Center (MSK) and the University of Pennsylvania (Penn). We considered any antibiotic exposure between day -30 and the day of CAR T cell infusion. We focused our analysis on anaerobe-targeting antibiotics used in the setting of neutropenic fever: piperacillin-tazobactam, imipenem-cilastatin, and meropenem (here referred to as "P-I-M"). We found that forty-seven (20.6%) of 228 patients were exposed to P-I-M in the four weeks before CAR T cell infusion. Patient characteristics at the time of CAR T cell infusion were similar between the P-I-M-exposed and not-exposed groups, although a worse performance status was observed in patients with NHL treated with P-I-M. We found that overall survival (OS) was significantly decreased following CAR T cell infusion in patients exposed to P-I-M (**Fig 1A**; OS HR, 2.58; 95% CI, 1.68 - 3.98; $p = <0.001$). A subgroup analysis of the patients with NHL also demonstrated decreased OS associated with P-I-M exposure whether the costimulatory domain was CD28 or 4-1BB (*data not shown*). Next, we queried whether patients exposed to P-I-M had more aggressive disease. We evaluated potential confounders for the findings in uni- and multi-variable models. Importantly, exposure to P-I-M remained a strong predictor of decreased OS (HR, 2.58; 95% CI, 1.55 - 4.3; $p = <0.001$) (**Table 1**). Exposure to P-I-M was also associated with increased ICANS ($p = 0.023$) but not CRS ($p = 0.058$) in patients in the combined NHL and ALL cohort as well as in patients with NHL (CRS: $p = 0.154$, ICANS: $p = 0.002$) (*data not shown*).

We also prospectively collected baseline fecal samples prior to cell infusion from CD19 CAR T cells recipients (n=48) at MSK and Penn. Samples were submitted for 16S RNA sequencing of the V4-V5 region on the Illumina MiSeq platform and the amplicon sequence variants (ASVs) were annotated according to the NCBI 16S database using BLAST. In comparison to healthy controls (n=30), we found that alpha-diversity was significantly lower in fecal samples from CAR T cell patients ($p = 0.0023$, **Fig 1B**) and the composition of fecal samples was significantly different ($p = <0.001$, **Fig 1C**). Finally, linear discriminant analysis effect size (LEfSe) identified an increased abundance of *Lachnospiraceae*, *Ruminococcaceae*, and *Bacteroidaceae* in patients who achieved a Day 100 complete response (CR) and those who experienced CAR-mediated toxicity (*data not shown*).

Conclusion: Our results suggest that exposure to antibiotics, in particular P-I-M, in the four weeks before therapy was associated with worse survival. Profiling of the baseline fecal microbiome samples by 16S revealed that CD19 CAR T cell patients presented with evidence of an altered fecal microbiome as measured by lower alpha-diversity and a composition that is distinct from that of healthy controls. Finally, we identified bacterial taxa that were associated with Day 100 CR and CAR-mediated toxicity. Our findings indicate that the intestinal microbiome can affect the efficacy of CD19 CAR T cell therapy and provides a rationale to target the intestinal microbiome to improve clinical outcomes of patients treated with cellular therapies.

Disclosures Smith: Janssen: Consultancy, Honoraria. **Gomes:** Xbiome: Current Employment. **Schluter:** Postbiotics Plus LLC: Other: cofounder. **Park:** Kura Oncology: Consultancy; BMS: Consultancy; Servier: Consultancy; Autolus: Consultancy; Curocel: Consultancy; Artiva: Consultancy; Kite Pharma: Consultancy; Amgen: Consultancy; Novartis: Consultancy; Affymune: Consultancy; Intellia: Consultancy; Innate Pharma: Consultancy; Minerva: Consultancy; PrecisionBio: Consultancy. **Palomba:** Pharmacyclis: Membership on an entity's Board of Directors or advisory committees; Kite Pharmaceuticals: Membership on an entity's Board of Directors or advisory committees. **Jain:** Targeted Healthcare Communications: Consultancy; Bristol Myers Squibb: Other: for advisory board participation; CareDx: Other: for advisory board participation; CTI Biopharma: Research Funding; Syneos Health: Research Funding. **Pennisi:** Gilead Sciences: Consultancy. **Perales:** Miltenyi Biotec: Honoraria, Other; Novartis: Honoraria, Other; Omeros: Honoraria; NexImmune: Honoraria; Bristol-Myers Squibb: Honoraria; Merck: Honoraria; Celgene: Honoraria; Takeda: Honoraria; Kite/Gilead: Honoraria, Other; Medigene: Honoraria; Nektar Therapeutics: Honoraria, Other; Cidara: Honoraria; Servier: Honoraria; Sellas Life Sciences: Honoraria; Karyopharm: Honoraria; MorphoSys: Honoraria; Equilium: Honoraria; Incyte: Honoraria, Other. **Garfall:** Amgen: Honoraria; CRISPR Therapeutics: Research Funding; GlaxoSmithKline: Honoraria; Janssen: Honoraria, Research Funding; Novartis: Research Funding; Tmunity: Research

	CD20	1.04 (0.63-1.73)			
Disease	ALL	(reference)	0.097	(reference)	0.130
	NHL	0.72 (0.48-1.06)		0.72 (0.46-1.1)	
Gender	Female	(reference)	0.157		
	Male	1.37 (0.89-2.1)			
LDH	Continuous, per 100	1.33 (1.18-1.5)	<0.001	1.34 (1.18-1.52)	<0.001

Table 1. Uni- and multivariable analysis of the association of patient characteristics and overall survival following CD19-targeted CAR T cell therapy, stratified by institution. Uni- and multivariable Cox proportional-hazards analyses of the association of decreased overall survival in patients who were exposed to piperacillin-tazobactam, imipenem-cilastatin or meropenem (P-I-M) in the 30 days prior to CD19-targeted CAR T cell therapy and patient characteristics. The statistical threshold for inclusion of a variable in the multivariate model is <0.10.

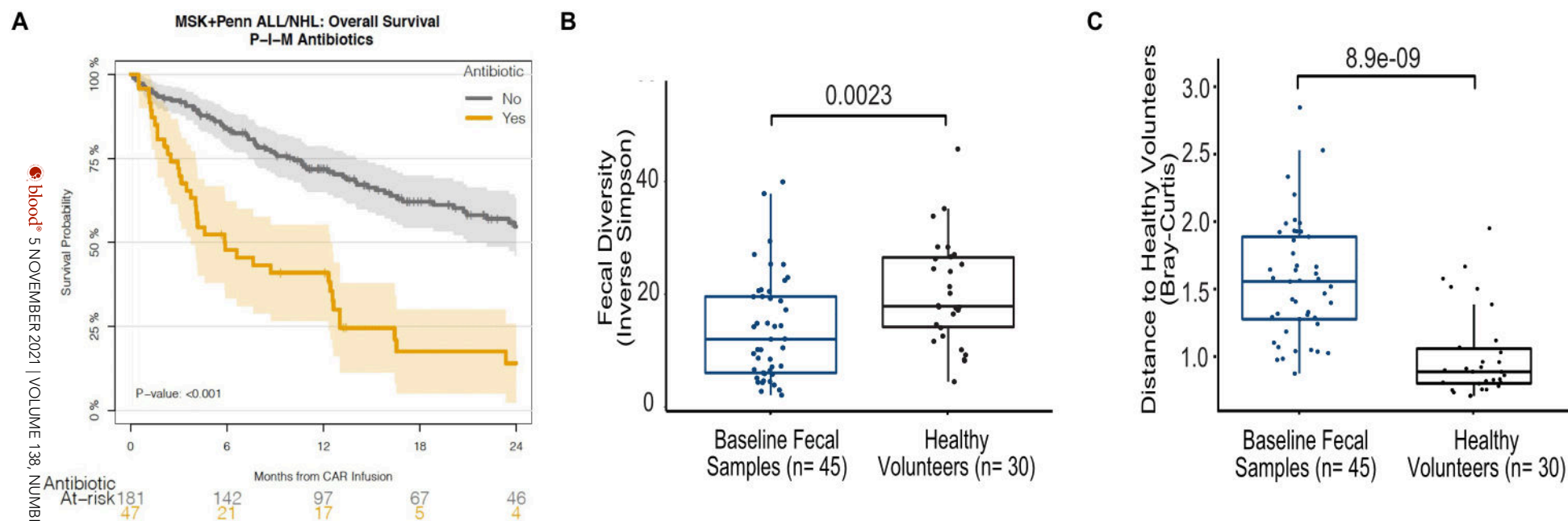


Figure 1. (A) Kaplan-Meier curve of overall survival by log-rank test according to exposure to P-I-M antibiotics the within 4 weeks before CD19 CAR T cell infusion in patients with ALL and NHL. (B) Inverse Simpson diversity index of the fecal microbiome in the baseline fecal samples (n= 45) and healthy volunteers (n= 30) by Wilcoxon rank-sum test. (C) Bray-Curtis beta-diversity distances between a reference point defined by the average of healthy volunteers and each of 30 samples from healthy volunteers and 45 baseline patient samples by Wilcoxon rank-sum test.

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